A Practical Introduction to Machine Learning in Python

Day 5 – Friday

»Transformers«

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Gesis

This part: State of the art and next steps

Recap

Word vectors

- Encoding text in ML models

Non-Contextual

Embeddings

- Using word embeddings to improve models

Contextual Embeddings

Transformer-based models

Transfer learning paradigm

- Do I need all this fancy stuff?

Ethical considerations

Your takeaway

Recap

How Humans Learn vs. How Machines Learn

Human Learning

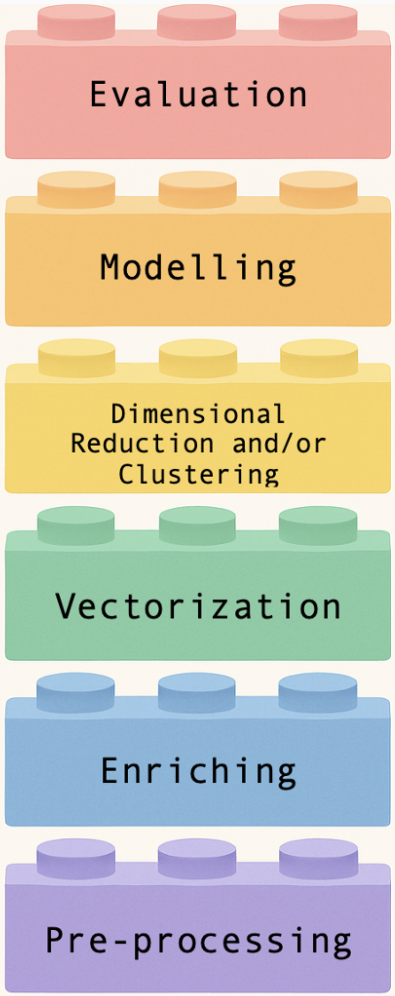
Example counting
(seeing many repetitions)

Pointing out
(explicit correction / feedback)

Explaining in other words
(paraphrase / abstraction)

Interpreting context
(discourse / world knowledge)

Lego pipeline



how to assess the output of your model

how to train a model to describe (UML) or to predict (SML)

how to group features *or* cases

how to turn text → numbers

how to label text data

how to tidy up text data

Timeline of Key Embedding Models

- **Word2Vec** – 2014
  Introduced efficient neural word embeddings.

- **Transformers** – 2017
  Vaswani et al. introduced the Transformer architecture, enabling powerful sequence modeling.

- **BERT** – 2018
  Devlin et al. introduced BERT, a bidirectional transformer for language understanding.

- **ChatGPT-3.5** – 2022
  OpenAI released ChatGPT-3.5, a large language model for conversational AI.

Word vectors

Our BOW approach until now

Representing a document by word frequency counts

Result of preprocessing and vectorizing:

0. `He took the dog for a walk to the dog playground`

⇒ `took dog walk dog playground`

⇒ `'took':1, 'dog': 2, walk: 1, playground: 1`

Consider these other sentences

1. He took the doberman for a walk to the dog playground
2. He took the cat for a walk to the dog playground
3. He killed the dog on his walk to the dog playground

The vectorized representations of these sentences have a "distance" (dissimilarity) of 1 each, but arguably, sentences 0 and 1 should be "closer" than others

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Word vectors

Encoding text in ML models

Encoding so far: BoW approach

- Our vectorizers gave a random ID to each word
- Words are *discrete* and *independent* tokens.
- This is a rather naïve assumption, with two main disadvantages:
  1. high dimensionality of the tokens
  2. they do not incorporate real-world knowledge

Token	Index	One-hot vector
aargh	0	[1,0,0]
king	1	[0,1,0]
queen	2	[0,0,1]

Non-Contextual Embeddings

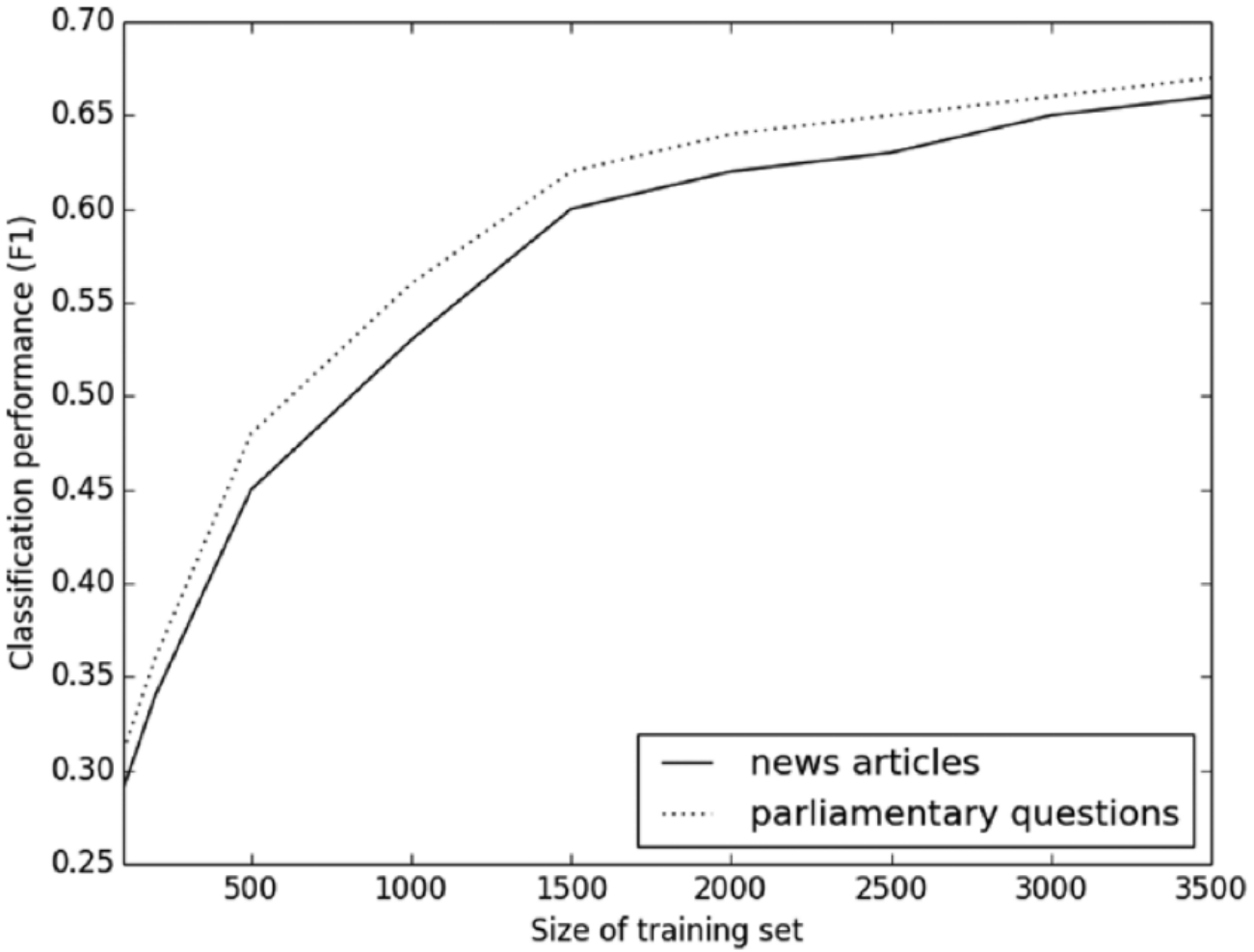
"...a word is characterized by the company it keeps..." (Firth, 1957)

Word embeddings . . .

- help capturing the meaning of text
- are low-dimensional vector representations that capture semantic meaning
- for instance, 'dobermann' and 'bulldog' should be represented by vectors that are close to each other in space, while 'kill' and 'walk' should be far from each other.

Training size vs performance Burscher et al., 2015

FIGURE 1
Learning Curves for the Classification of News Articles and PQs



Word embeddings:Training algorithms

There were two popular approaches to training śtaticẃord embeddings: GloVe and word2vec.

- GloVe is count-based: dimensionality reduction on the co-occurrence counts matrix.
- Word2Vec is a predictive model: neural network to predict words/contexts
- That means that GloVe takes global context into account, word2vec local context
- Some technical implications for how training can be implemented
- **However, only subtle differences in final result.**

Word2Vec: Continous Bag of Words (CBOW) vs skipgram

Example sentence: "the quick brown fox jumped over the lazy dog"

CBOW: Predict a word given the words in its context

Dataset:

```
([the, brown], quick), ([quick, fox], brown),
([brown, jumped], fox), ...
```

skipgram: Predict the context words given the target word

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(quick, the), (quick, brown), (brown, quick), (brown,
fox), ...
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Example taken from here: https://medium.com/explore-artificial-intelligence/word2vec-a-baby-step-in-deep-learning-but-a-giant-leap-towards-natural-language-processing-40fe4e8602ba

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From static to contextual embeddings

- **Word2Vec:**
  - Learns a single fixed vector for each word.
  - Embeddings are based on neighboring words.
  - Cannot distinguish between different meanings of a word in different sentences.

- **Transformers:**
  - Dynamically compute word representations based on the entire sentence.
  - Each word's embedding can change depending on surrounding words.
  - Attention mechanism allows the model to focus on relevant words anywhere in the input.

Training contextual word embeddings

- Language modelling is a set of techniques that aim to determine the probability of a given sequence of words occurring in a sentence.
- Large language Models: Language modelling applied to massive amounts of training data (e.g., Wikipedia, news archives, Reddit, etc.)
- Often referred to as 'pretrained' or 'foundation' models.

Training language models: based on unstructured text – in the absence of explicit labels

Dobermann

From Wikipedia, the free encyclopedia

"Doberman" redirects here. For other uses, see Doberman (disambiguation).

This article **needs additional citations for verification**. Please help improve this article by adding citations to reliable sources. Unsourced material may be challenged and removed.

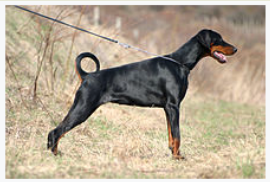
Find sources: "Dobermann" – news · newspapers · books · scholar · JSTOR

(March 2018) (Learn how and when to remove this template message)

The **Dobermann** (/ˈdoʊbərmən/; German pronunciation: [ˈdoːbɐman]), or **Doberman Pinscher** in the United States and Canada, is a medium-large breed of domestic dog that was originally developed around 1890 by Louis Dobermann, a tax collector from Germany.[2] The Dobermann has a long muzzle. It stands on its pads and is not usually heavy-footed. Ideally, they have an even and graceful gait. Traditionally, the ears are cropped and posted and the tail is docked. However, in some countries, these procedures are now illegal and it is often considered cruel and unnecessary. Dobermanns have markings on the chest, paws/legs, muzzle, above the eyes, and underneath the tail.

Dobermanns are known to be intelligent, alert, and

Dobermann



Other names	Doberman Pinscher
Common nicknames	Dobie, Doberman
Origin	Germany

Traits [hide]

Height	Dogs	68 to 72 cm (27 to 28 in)[1]
	Bitches	63 to 68 cm (25 to 27 in)[1]

Bulldog

From Wikipedia, the free encyclopedia

This article is about the English Bulldog. For other uses, see Bulldog (disambiguation).
See also: French Bulldog, American Bulldog, and Old English Bulldog

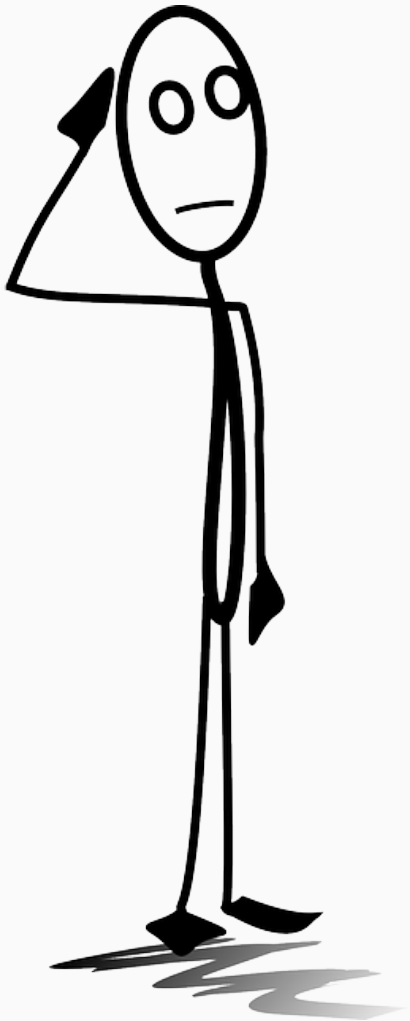
The **Bulldog** is a British breed of dog of mastiff type. It may also be known as the **English Bulldog** or **British Bulldog**. It is of medium size, a muscular, hefty dog with a wrinkled face and a distinctive pushed-in nose.[4] It is commonly kept as a companion dog; in 2013 it was in twelfth place on a list of the breeds most frequently registered worldwide.[5]

The Bulldog has a longstanding association with British culture; the BBC wrote: "to many the Bulldog is a national icon, symbolising pluck and determination".[6] During the Second World War, the Prime Minister Winston Churchill was likened to a Bulldog for his defiance of Nazi Germany.[7] The Bulldog Club (In England) was formed in 1878, and the Bulldog Club of America was formed in 1890.

Bulldog

Male English Bulldog

Other names	English Bulldog, British Bulldog
Origin	England[1]



What can we use word embeddings for?

Non-Contextual Embeddings

Improving down-stream classification tasks

In supervised machine learning

- Modify CountVectorizer or TfidfVectorizer:
- For each document, we look up the embedding of each word in a (often) pre-trained embedding model (e.g., trained on the whole Wikipedia)
- We then aggregate these embeddings (e.g., mean or sum)
- For each document, we now have a, for example, 300-dimensional instead of 10,000 dimesional vector.
- Use these vectors to predict the label.

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In supervised machine learning

What does this mean?

- Our model is smaller
- We can use words in the prediction set *even if they are not in the training dataset* (as long as it is in the embedding model!)
- We can learn from similar training samples even if they do not use the same exact words
- But we may also loose some nuance

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In document similarity calculation

Use cases

- plagiarism detection
- Are press releases/news agency copy/. . . taken over?
- Event detection

Traditional measures

- Levenshtein distance (How many characters|words do I need to change to transform string A into string B?)
- Cosine similarity ("correlation" between the BOW-representations of string A and string B)

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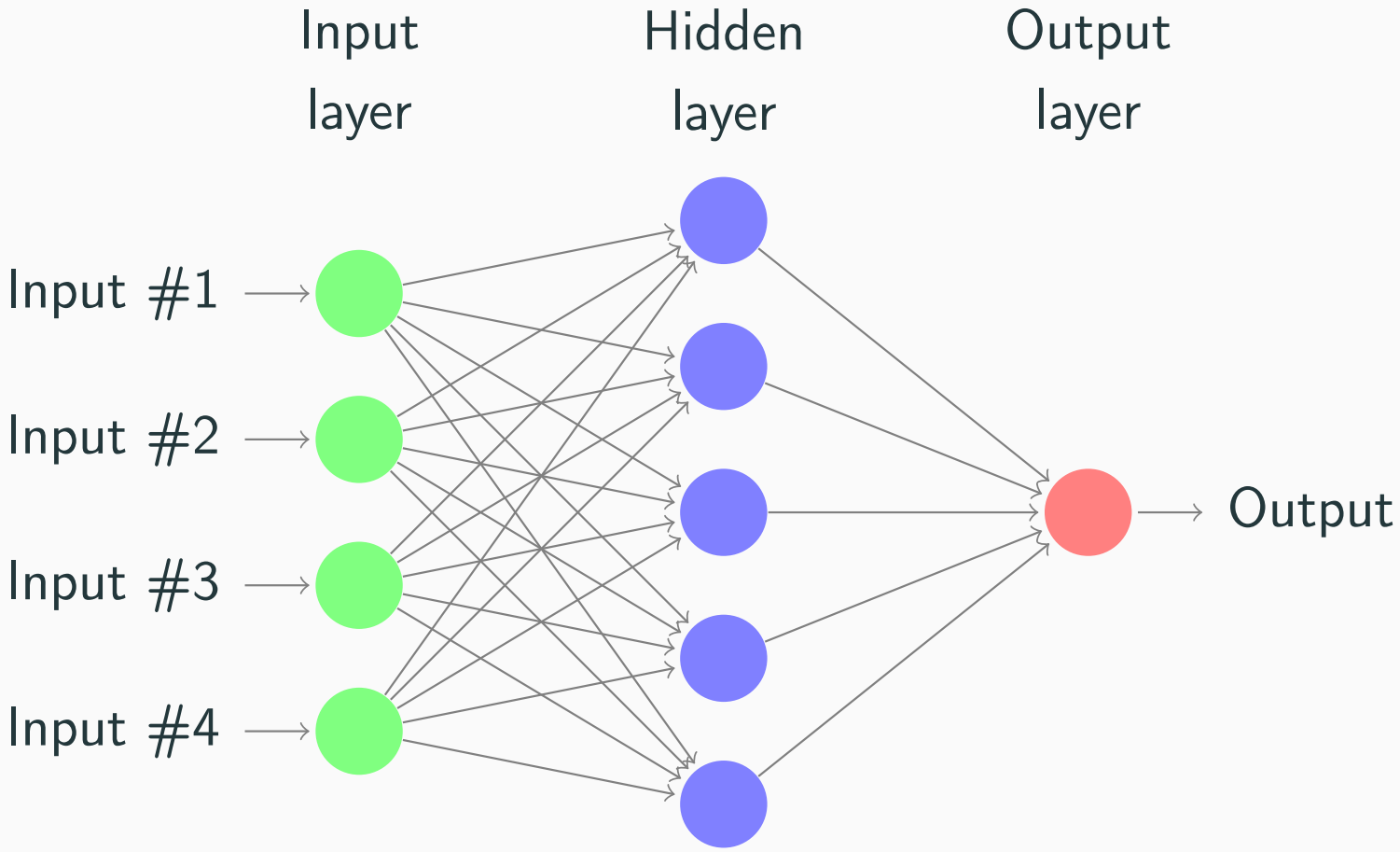
Example Visualization

Ongoing work within the TWON project, visualization credit to Simon M"unker, Trier University

Neural Networks

- In "classical" machine learning, we predict an outcome directly based on the input features
- In neural networks, we can have "hidden layers" that we predict
- These layers are not necessarily interpretable
- "Neurons" that "fire" based on an "activation function"



$\Rightarrow$ If we had multiple hidden layers in a row, we'd call it a *deep* network.

Why neural networks?

- learn hidden structures (e.g., embeddings (!))
- go beyond the idea that there is a direct relationship between occurrence of word X and label (or occurrence of pixel [R,G,B] and a label)
- images, machine translation — and more and more general NLP, sentiment analysis, etc.

Example of a comparatively easy introduction:
https://towardsdatascience.com/
neural-network-embeddings-explained-4d028e6f0526

Convolutional networks

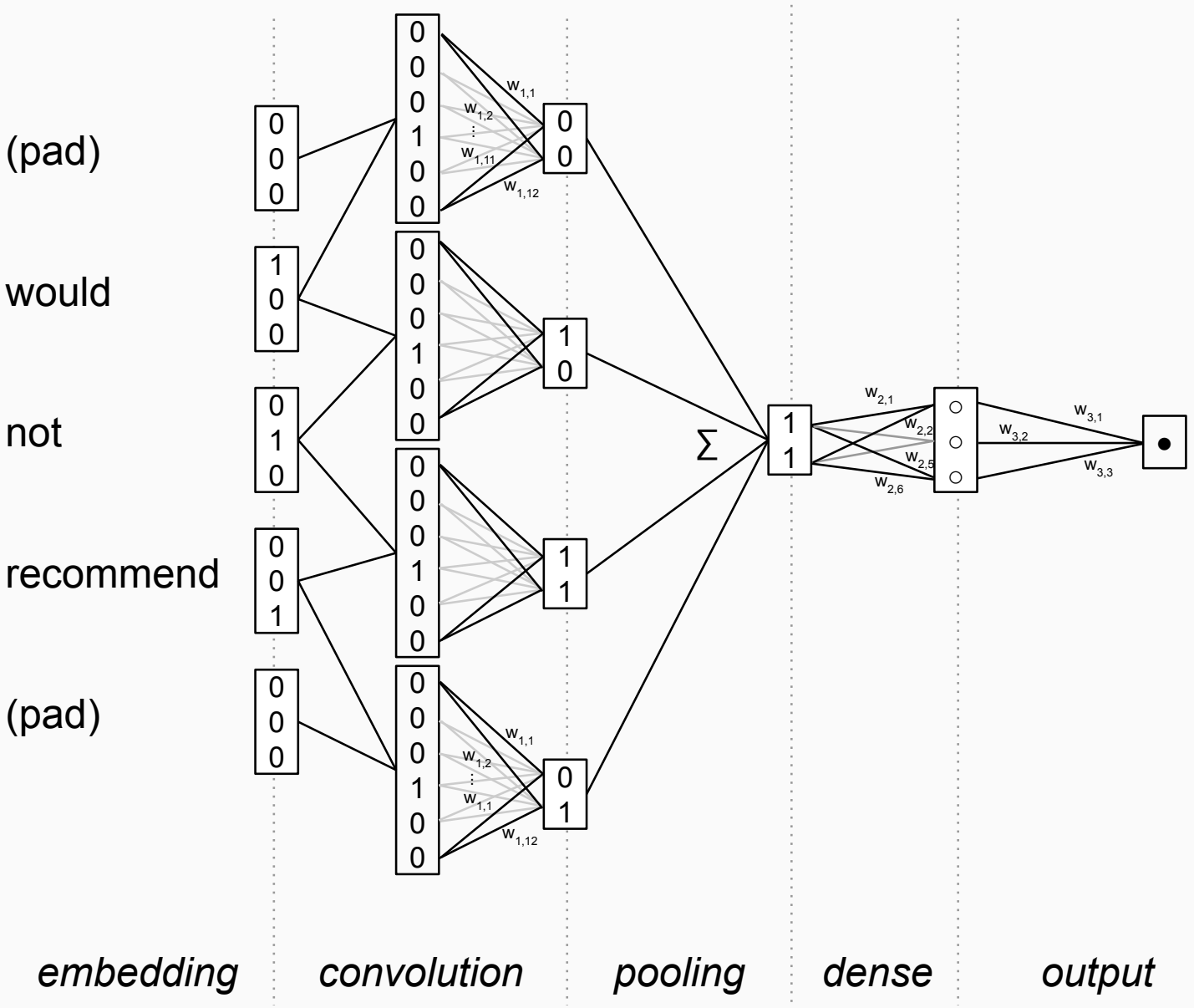
The problem with such a basic network: just as with classic SML, we still lose all information about order (the "not good" problem).

Therefore,

- We concatenate the vectors of neighboring words
- We apply some filter (essentially, we detect patterns)
- and then pool the results (e.g., taking the maximum)

This means that we now explicitly take into account *the temporal structure* of a sentence.

Convolutional networks



Contextual Embeddings

Downsides of Non-Contextual Word Embeddings

- Word2Vec and Glove produce *static* vectors: each word is represented by a single vector.
  - e.g., the vector for *date* is always the same...
  - ...however, the *meaning* of this word differs across domains: "she put a *date* in his lunchbox" (1); "they went on a *date*" (2); and "what's the *date* today?"

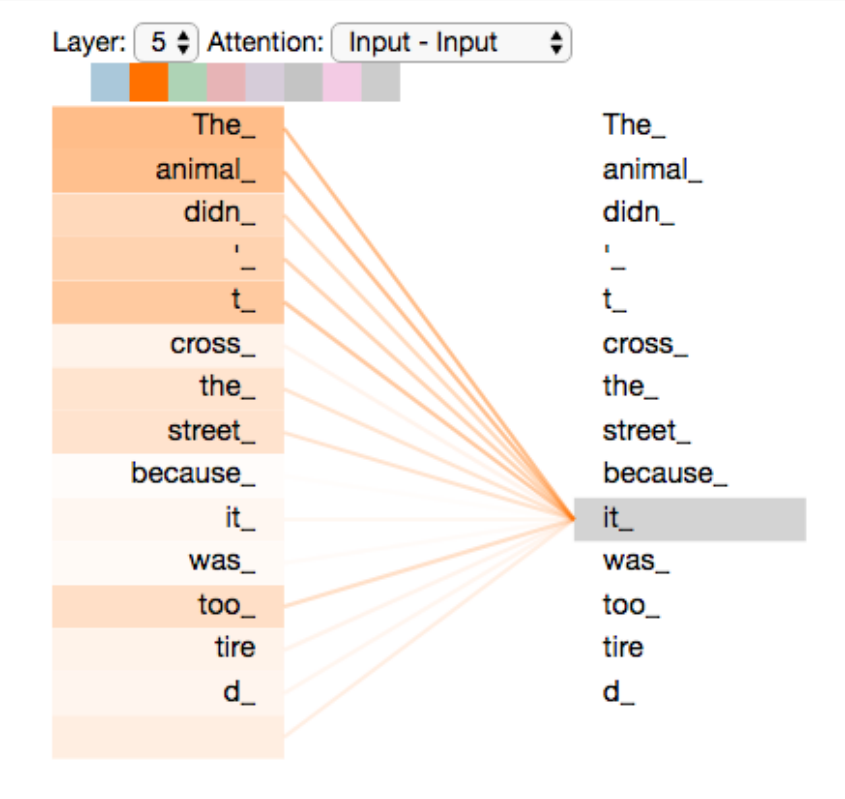
Enter: Contextual Word Embeddings

- *Transformers* create a new vector for each time a word is used in the dataset
- *Contextualized* vectors.
- *self-attention* mechanism is essential here: this is a manner to automatically decide which nearby words should influence a token's representation; the model *learns* which tokens *to attend to*

Transformer-based models

Attention Mechanism

"The animal didn't cross the street because it was too tired"



Transformer-based models

BERT: Bidirectional Encoder Representations from Transformers (Devlin et al., 2019)

- (Huge) pre-trained model (by, e.g., Google)
- Trained on very large amount of text and can use words in context.
- BERT and other transformer-based models are used for a range of tasks;
  1. Sentiment analysis (+/-)
  2. sequence-to-sequence predictions (e.g., translation)
  3. similarity and paraphrasing tasks
  4. natural language inference tasks (token prediction)

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How to use

HuggingFace

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Transfer learning paradigm

The idea of transfer learning is very powerful

Transfer Learning paradigm

1. Pre-train a model on data that is at hand (e.g., Wikipedia, Google News)
2. Fine-tune the model on your downstream task (bring in your small-scale annotated dataset)

By adding 'task-specific' heads you can produce specific outputs, e.g., classification, text generation, named entity recognition, etc.

Let's look at an example (exercises/transformers-custom-dataset.ipynb)

Transfer learning paradigm

Do I need all this fancy stuff?

Things to consider

How important is. . .

- precision/recall? Am I satisfied with .88 when .90 is theoretically possible? .85? .80? .75?
- explainability?
- computational resources?
- generalizability and out-of-sample performance?

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- Invest in good hyperparameter-tuning (cross-validation, gridsearch) and don't forget to set aside unseen data for the *final* evaluation.
- If you (a) need to get the highest possible accuracy, or (b) have reasons to assume that the model does not generalize well enough (overfitting problems, bad out-of-sample prediction (e.g., training topics on newspaper 1, predicting topics in newspaper 2), try embedding-based approaches, transformers, etc.
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Ethical considerations

(Ab-)using word embeddings to detect biases

Biased embeddings

- word embeddings are trained on large corpora
- As the task is to learn how to predict a word from its context (CBOW) or vice versa (skip-gram), biased texts produce biased embeddings
- If in the training corpus, the words “man” and “computer programmer” are used in the same context, then we will learn such a gender bias

Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings, 1–25. Retrieved from http://arxiv.org/abs/1607.06520

Biased embeddings

Usually, we do not want that (and it has a huge potential for a shitstorm)

unless. . .

we actually want to chart such biases.

Biased embeddings

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Biased embeddings

Usually, we do not want that (and it has a huge potential for a shitstorm)

unless. . .

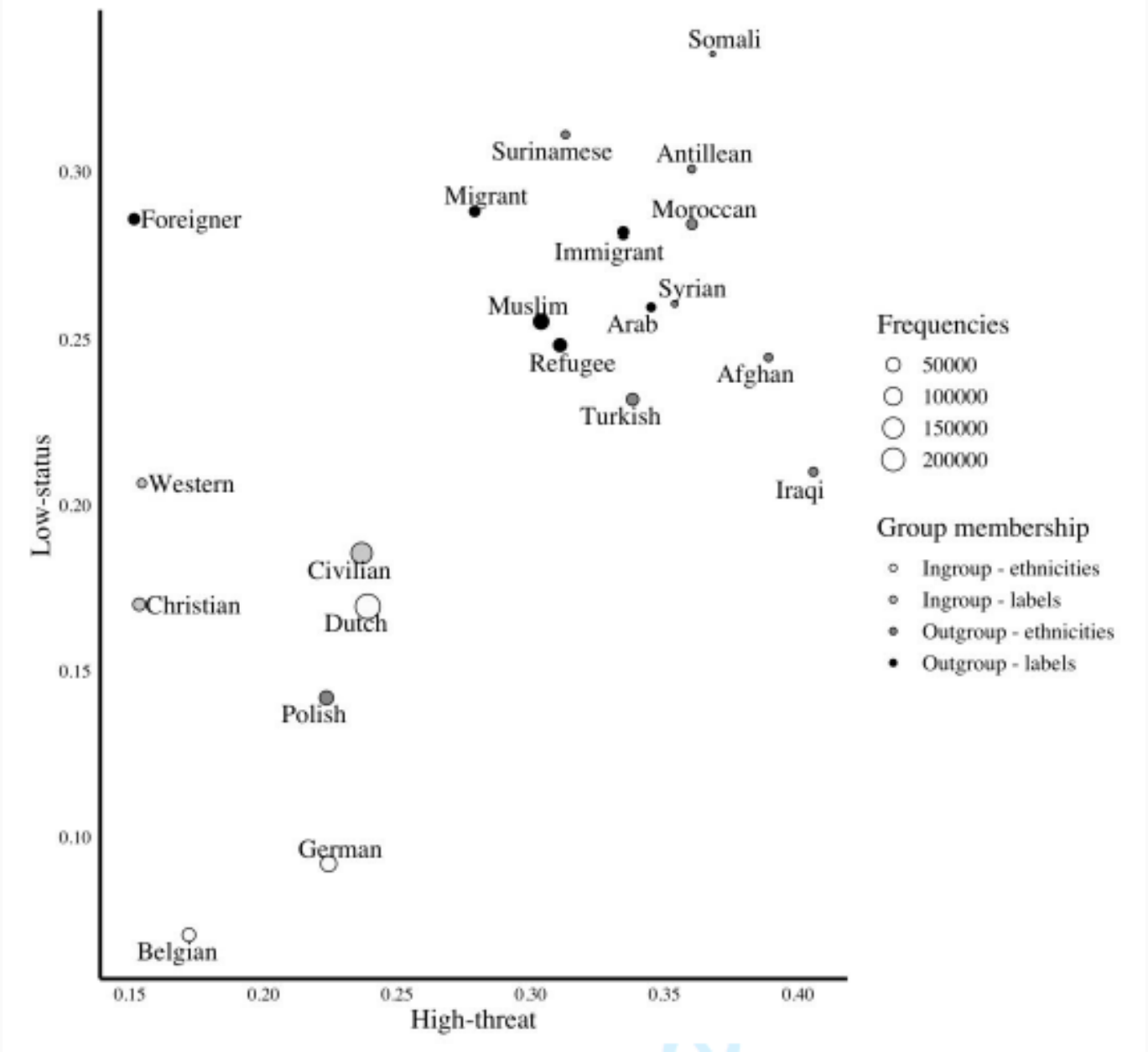
we actually want to chart such biases.

An example from our colleagues research

We trained word embeddings on 3.3 million Dutch news articles.

Are vector representations of outgroups (Maroccans, Muslims) closer to representations of negative stereotype words than ingroups?

Kroon, A.C., Van der Meer, G.L.A., Jonkman, J.G.F., &Trilling, D. (2021): Guilty by Association: Using Word Embeddings to Measure Ethnic Stereotypes in News Coverage. *Journalism & Mass Communication Quarterly*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Syrian
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Afghan
Turkish
Iraqi
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Civilian
Christian
Dutch
Polish
German
Belgian
Low-status
High-threat
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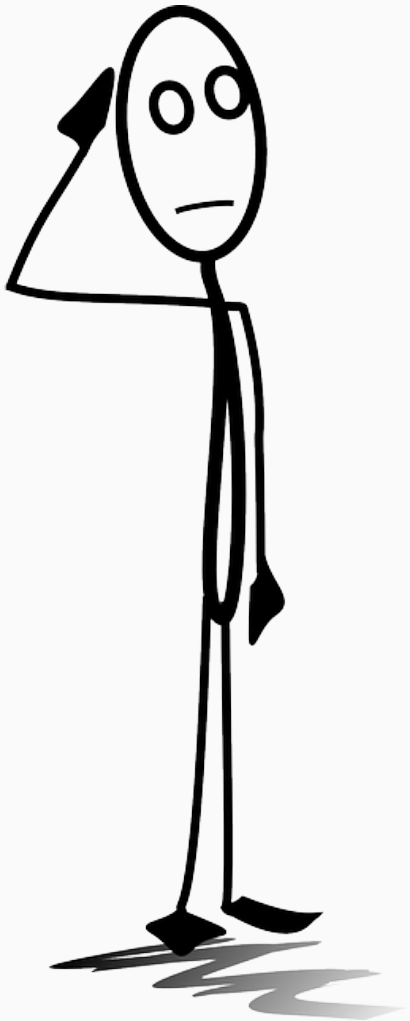
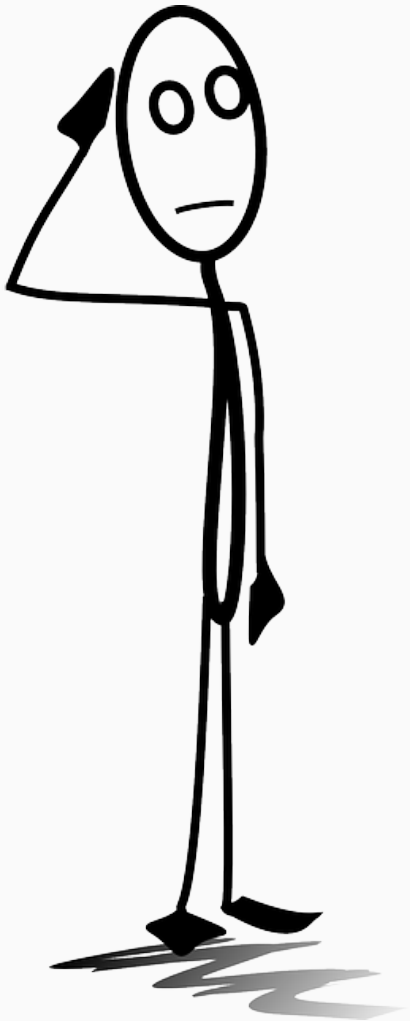
200000
Group membership
Ingroup - ethnicities
Ingroup - labels
Outgroup - ethnicities
Outgroup - labels

Your takeaway

(short recap of course)



Have your plans about how to and wether to use ML changed?



What are your next steps?

Last part: we help you working on (or discussing about) your own projects.

Training data

Training data set

- Dataset: diverse print and online news sources
- Preprocessing: duplicate sentences were removed
- Telegraaf (print & online), NRC Handelsblad (print & online), Volkskrant (print & online), Algemeen Dabldad (print & online), Trouw (print & online), nu.nl , nos.nl
- # words: 1.18b (1181701742)
- # sentences: 77.1M (77151321)

Training model

Training model

- We trained the model using Gensim's Word2Vec package in Python
- Skip-gram with negative sampling, window size of 5, 300-dimensional word vectors

Evaluation

Evaluation of the Amsterdam Embedding Model

Evaluation

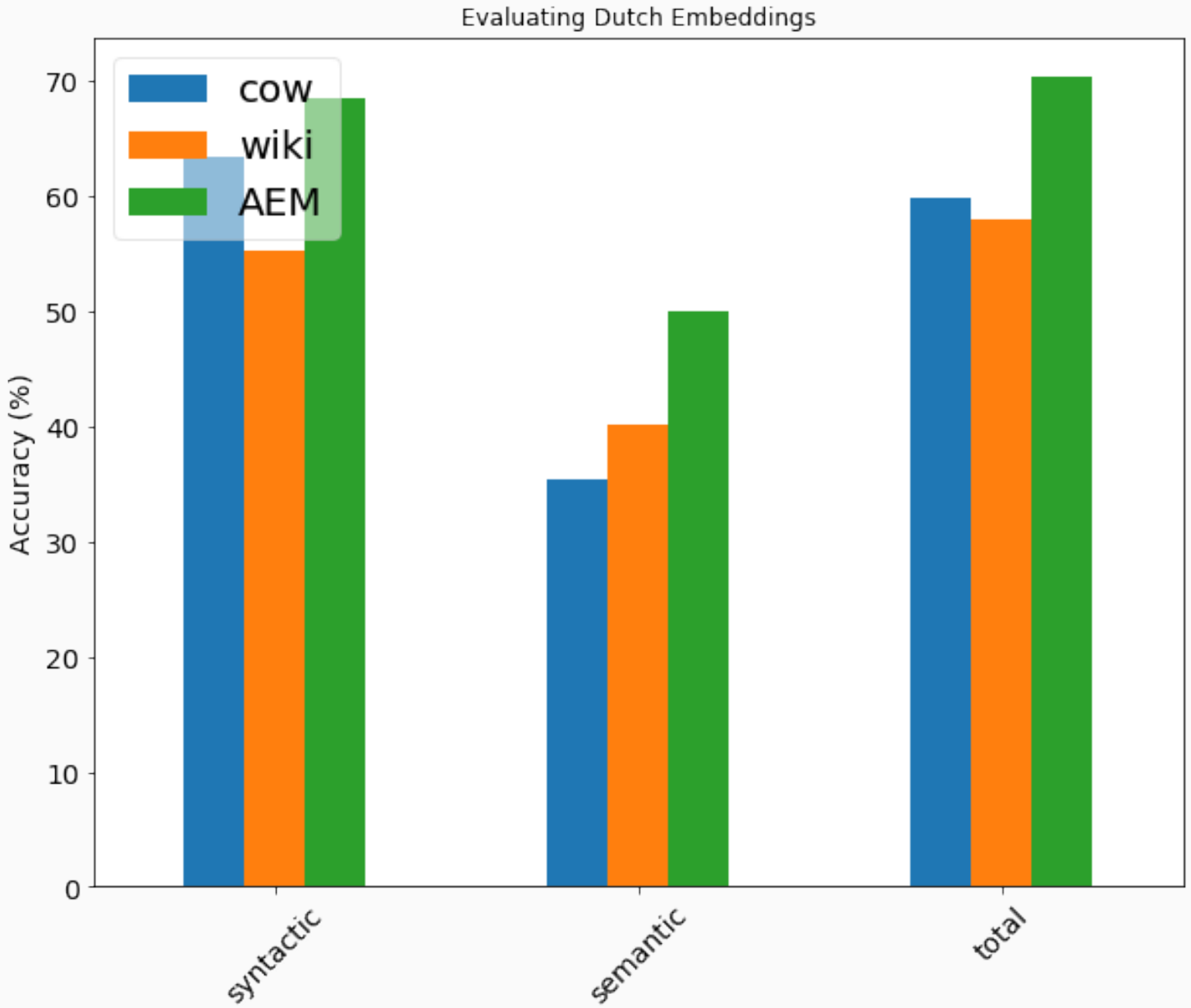
Evaluation methods

- To evaluate the model, we compare it to two other publicly available embedding models
  - $\Rightarrow$ **'Wiki'**: Embedding model trained on Wikipedia data (FastText)
  - $\Rightarrow$ **'Cow'**: Embedding model trained on diverse .nl and .be sites (Schafer & Bildhauer, 2012; Tulkens et al., 2016)
  - $\Rightarrow$ **'AEM'**: Amsterdam Embedding Model

Evaluation data

Evaluation data

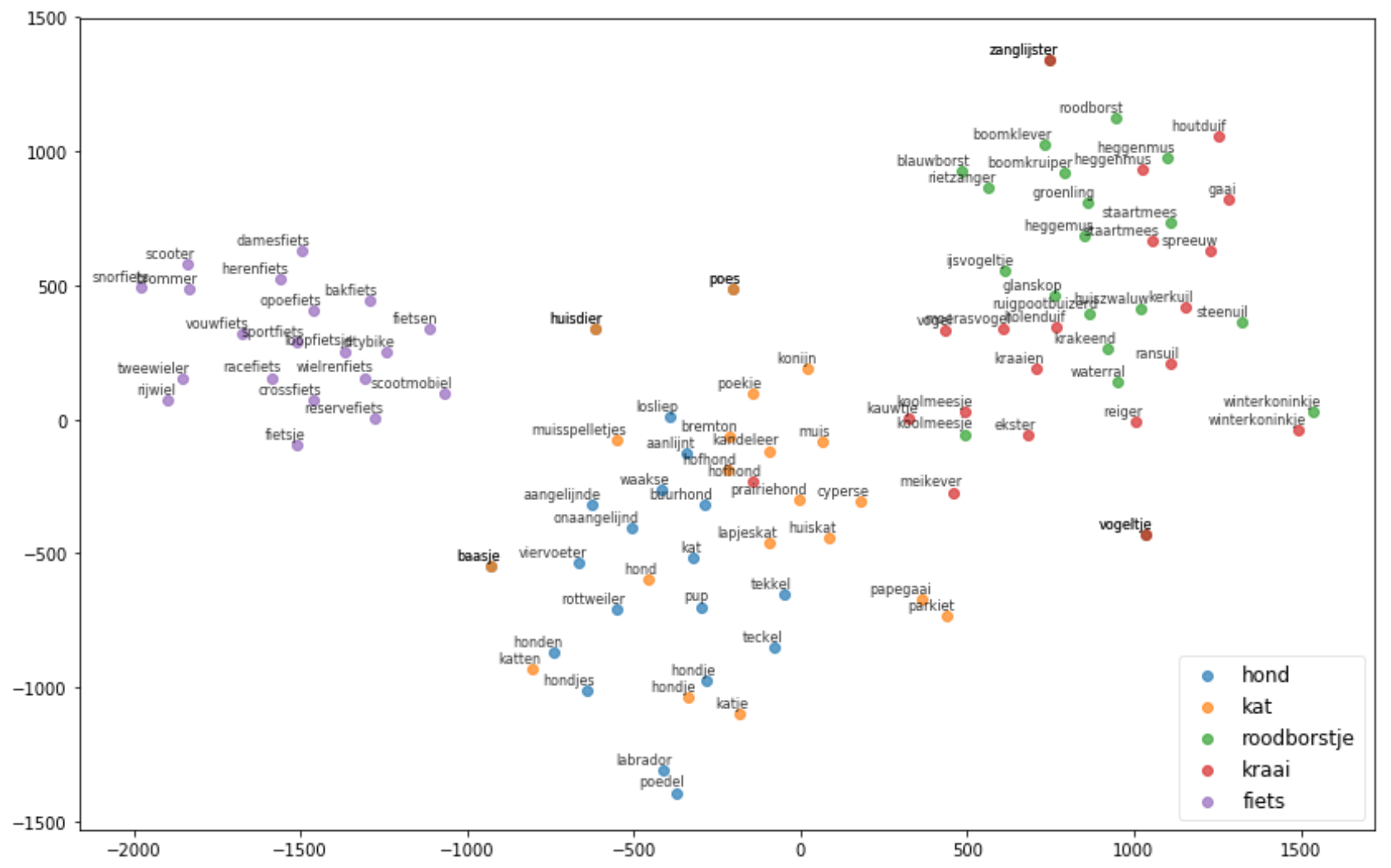
1. 'relationship'/ analogy-task (Tulkens et al., 2016)
   - **syntatic relationships**: dans dansen loop *[lopen]*
   - **semantic relationships**: denemarken kopenhagen noorwegen *[oslo]*
2. 5806 relationship tasks


Evaluating Dutch Embeddings
cow
wiki
AEM
Accuracy (%)
0
10
20
30
40
50
60
70
syntactic
semantic
total

Illustration

Illustration - Using the Amsterdam Embedding Model

Your takeaway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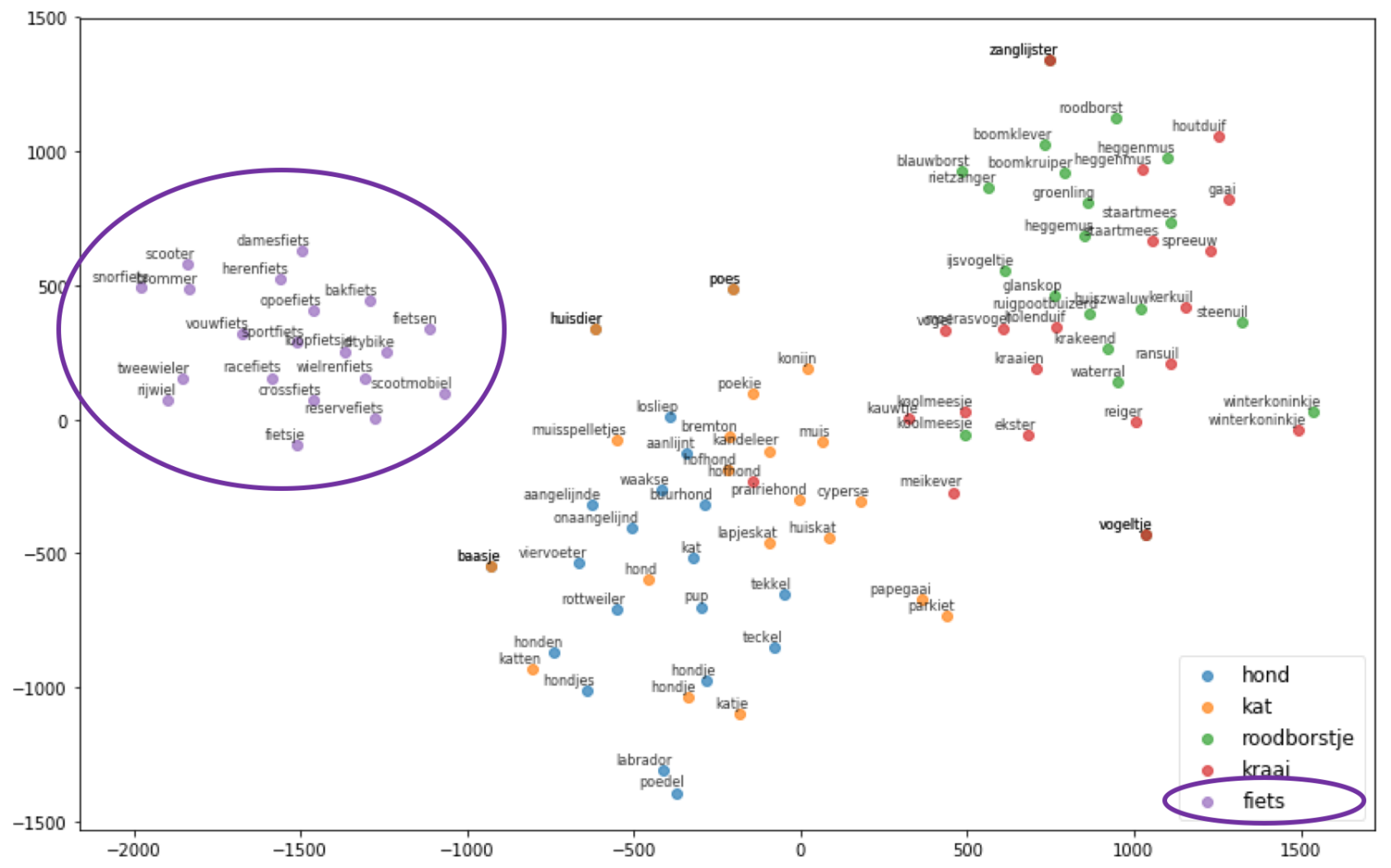
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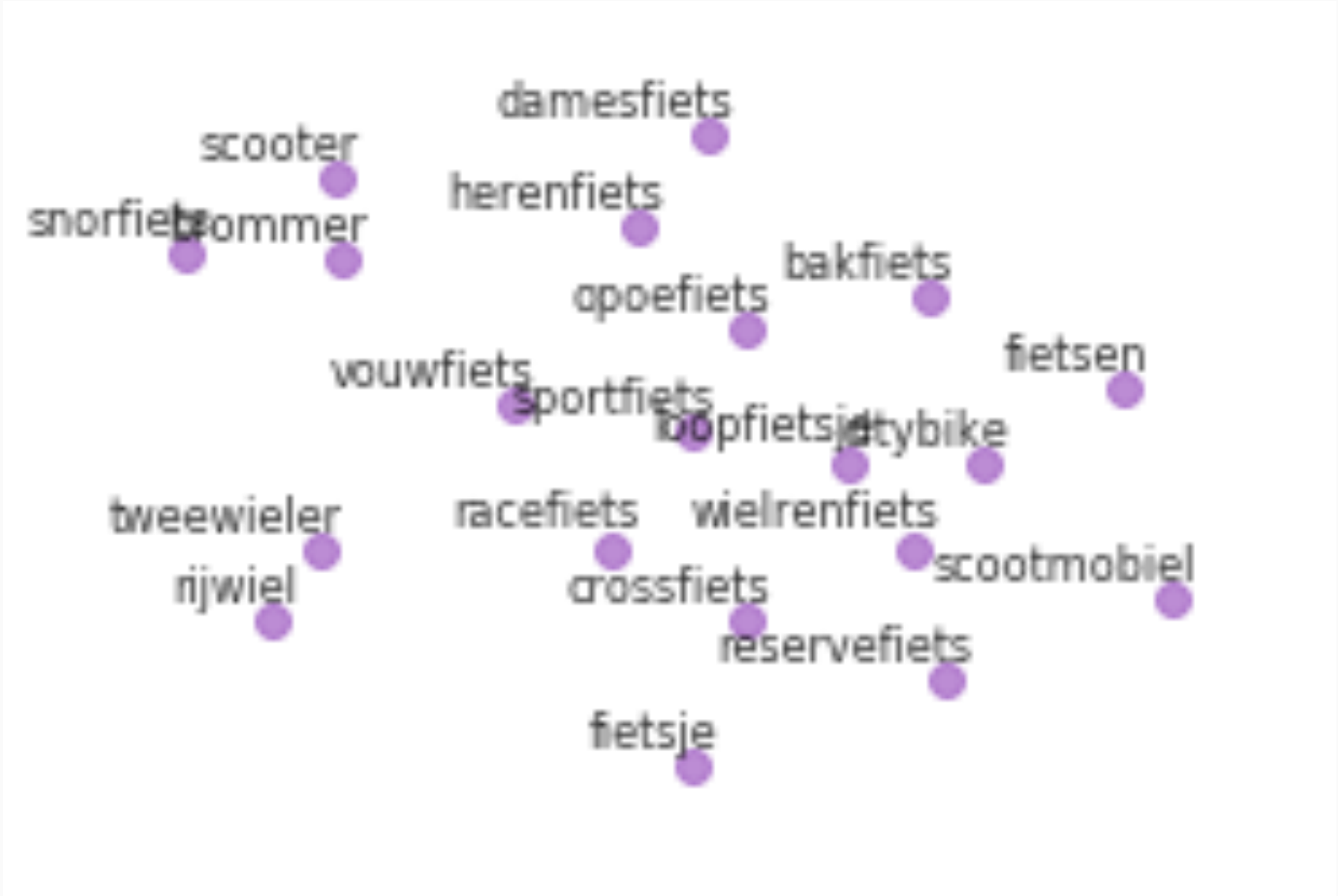
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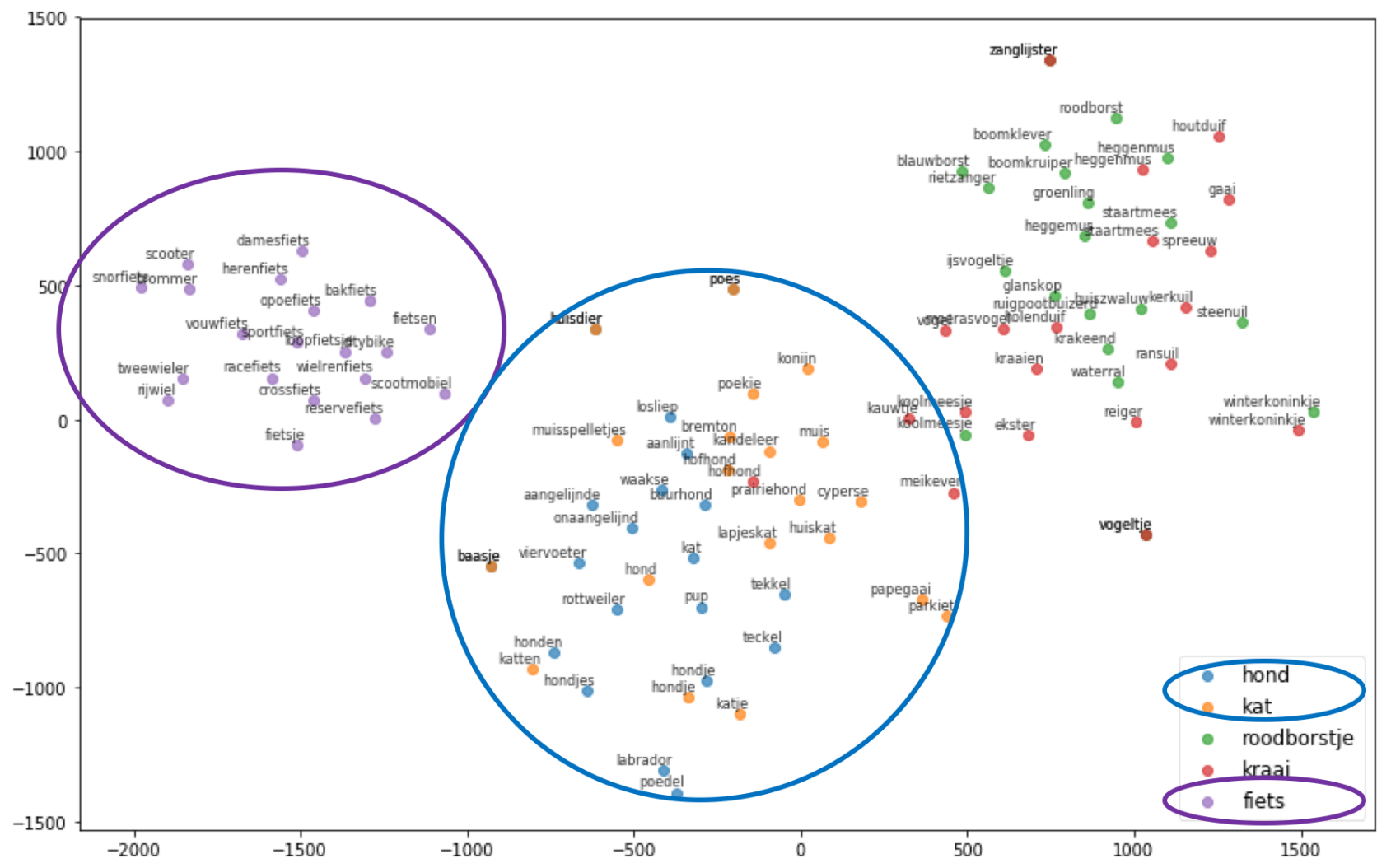
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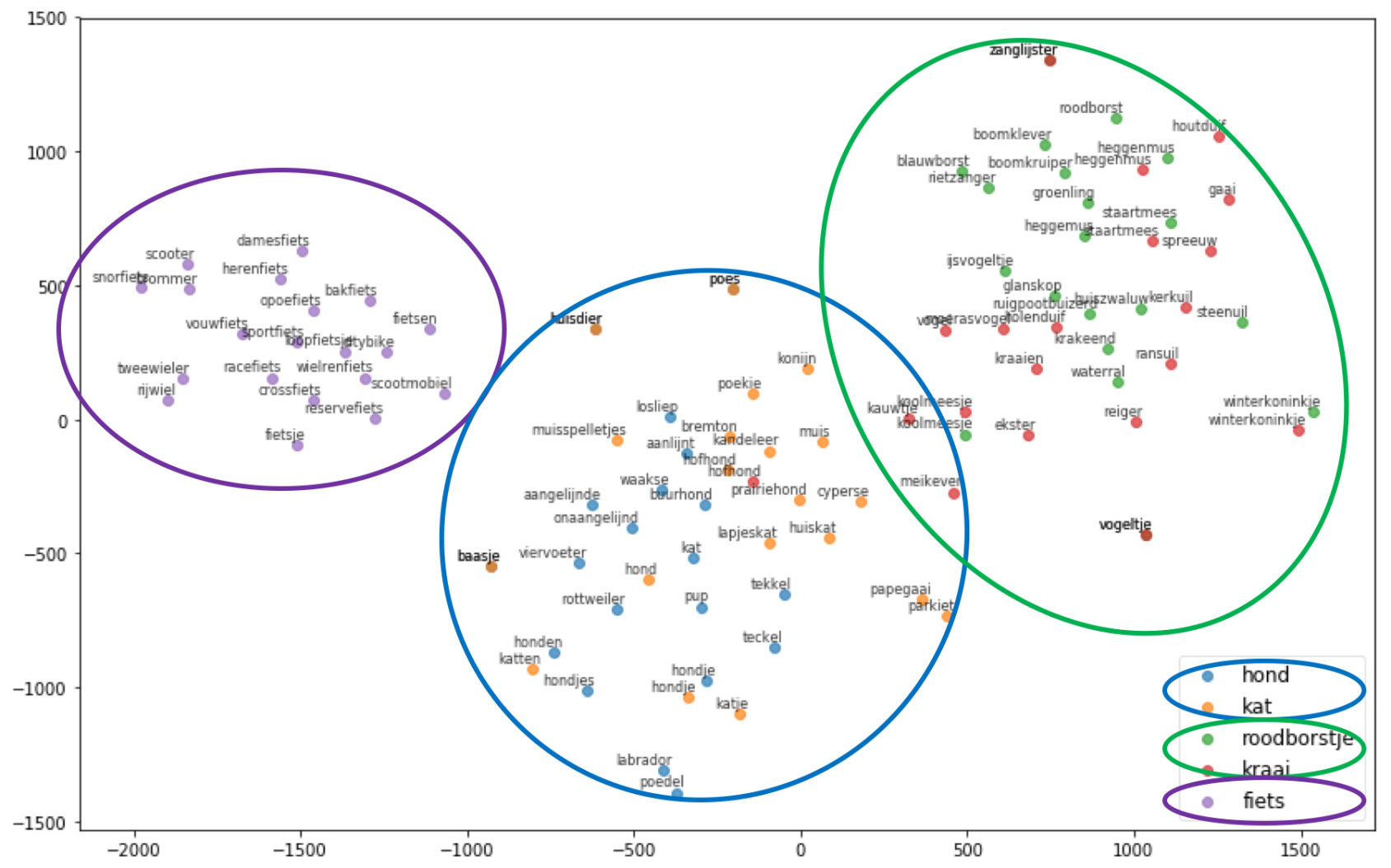
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Re-usability

Re-usability of the Amsterdam Embedding Model

Re-usability

Reusing model and data

1. See
   https://github.com/annekroon/amsterdam-embedding-model
2. Open access to all the code

Disclaimer: I cannot give a full overview of the whole topic of deep learning here – that's a whole (extensive) course in itself. But embeddings are closely related, that's why we at least will at least get out feet wet a bit.

References


Burscher, B., Vliegenthart, R., & De Vreese, C. H. (2015).**Using supervised machine learning to code policy issues: Can classifiers generalize across contexts?** *The ANNALS of the American Academy of Political and Social Science*, *659*(1), 122–131. https://doi.org/10.1177/0002716215569441


Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019).**BERT: Pre-training of deep bidirectional transformers for language understanding.** *NAACL HLT 2019 - 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Proceedings of the Conference*, *1*(1950), 4171–4186.